



Fruit Recognition for Efficient Sorting & Harvesting

Presented by:

Alorro ● Bocala ● Dañocup ● Maravilla ● Senibalo

start harvesting →



FRESH is a fruit
classification system for
agricultural applications.





Improves efficiency in
sorting fruits into fresh and
rotten categoriestion system
for agricultural applications.





Reduces reliance on
manual inspection into
postharvest handling.





Uses CNN to analyze fruit images based on color, shape, and texture.





Aligned with deep learning
approaches in agricultural
fruit quality classification
[1].



PROBLEM STATEMENT



Postharvest losses remain high due to inefficiencies in fruit quality handling.

Manual inspection of fruits—fresh vs. rotten—is slow and labor-intensive

Human-based grading is inconsistent and prone to missed defects or subjective judgment.

There is a need for an automated system that accurately classifies fruit quality using image-based analysis.



METHOD AND TECHNIQUES USED

Step 1 Data Acquisition & Wrangling



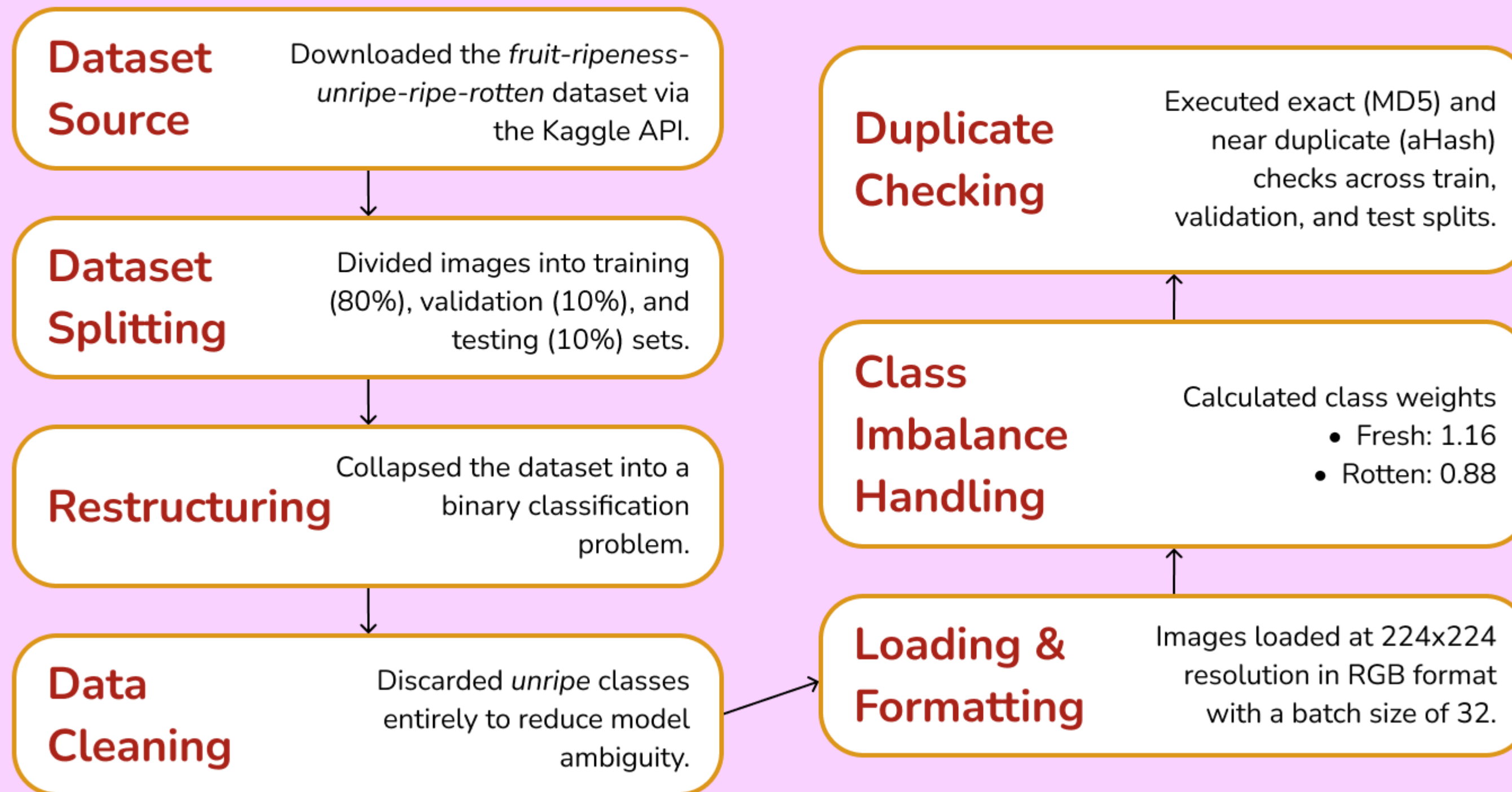
Step 2 & 3 Model Architecture
Comparison



Step 4 The Training Process



Data Acquisition & Wrangling



Model Architecture Comparison ✖

Feature

Approach

Custom CNN

Built from
scratch
(Sequential)

MobileNetV2

Transfer
Learning
(Pre-trained)

Feature

Extraction

3 Custom

Conv2D layers

Frozen

MobileNetV2

Model Architecture Comparison ✖

Feature
Extraction

3 Custom
Conv2D layers
(32, 64, 128
filters)

Frozen
MobileNetV2
base (Learned
from
Imagenet)

Classifier

Flatten →

Global

Model Architecture Comparison



Classifier
Head



Output



Flatten →
Dropout →
Dense
(128 units)



Dense



Global
Average
Pooling →
Dense (64) →
Dropout



Dense

Model Architecture Comparison ✖

Output
Layer

Dense
(2 units : Fresh
vs Rotten)

Dense
(2 units : Fresh
vs Rotten)

Trainable
Parameters

~11.1 Million
(Computationally
heavy)

~82,000
(Lightweight &
Fast)

RESULTS & VISUALIZATION

Training Performance



Model Evaluation



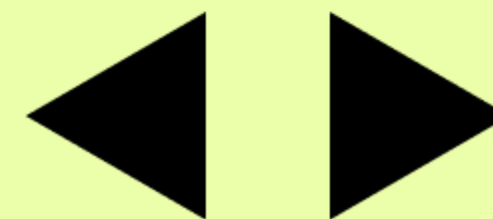
Visual Sanity Check



Real-World Testing



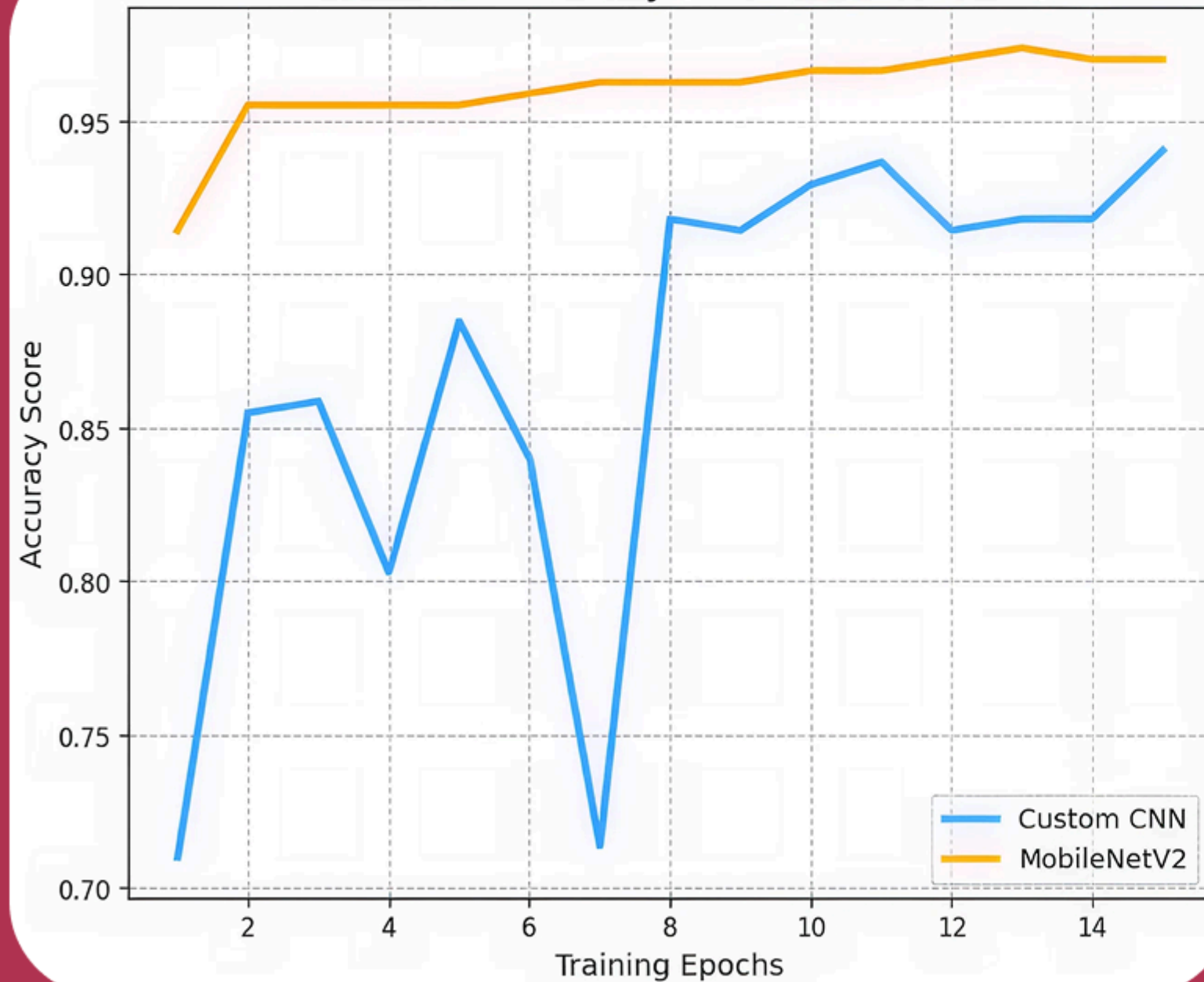
Grad-CAM Visualization



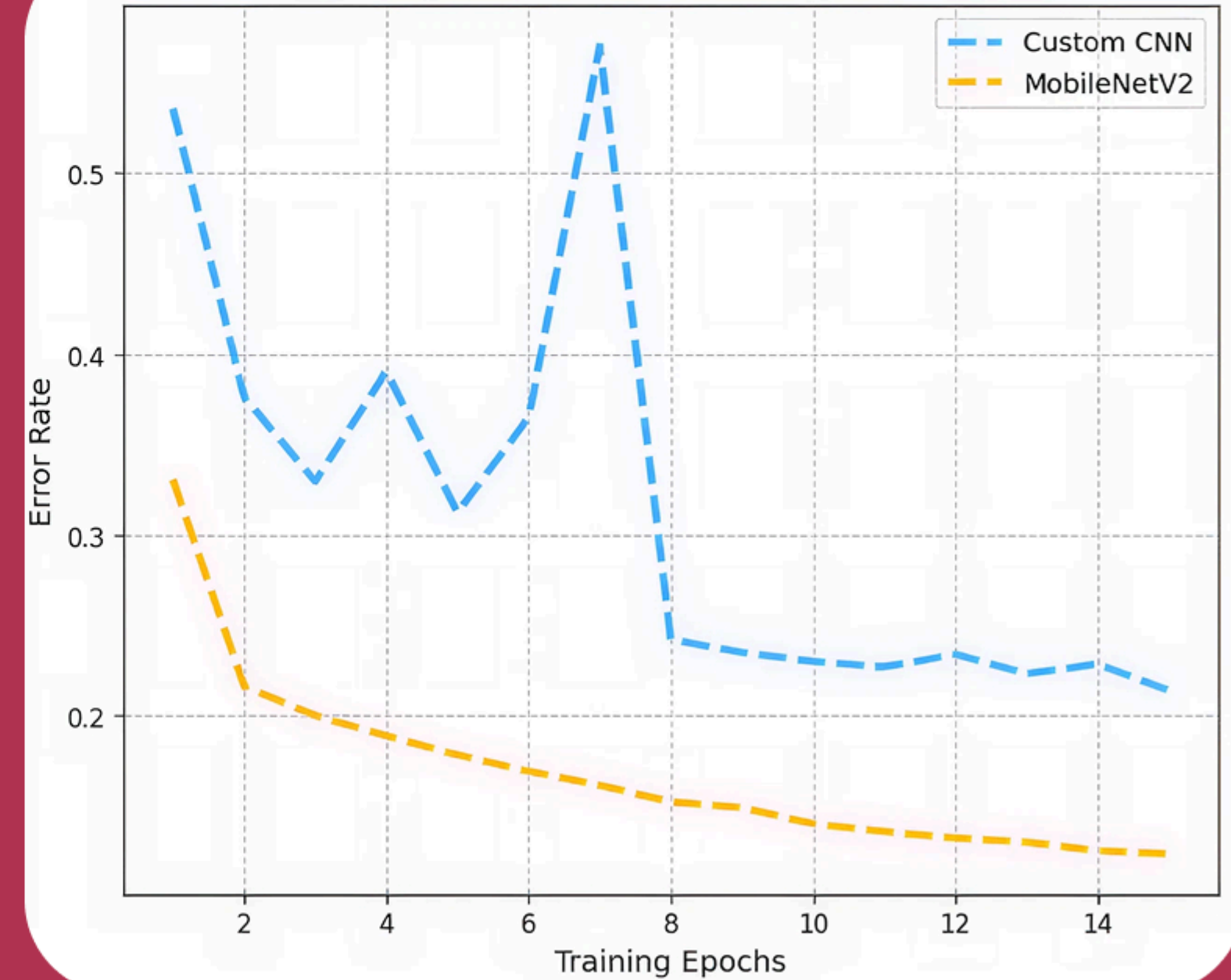
Training Performance



Validation Accuracy (The Race to 100%)



Validation Loss (Fewer Mistakes = Better)



Model Evaluation



✓ Confusion Matrix (Custom CNN):

```
[[108  10]
 [  9 146]]
```

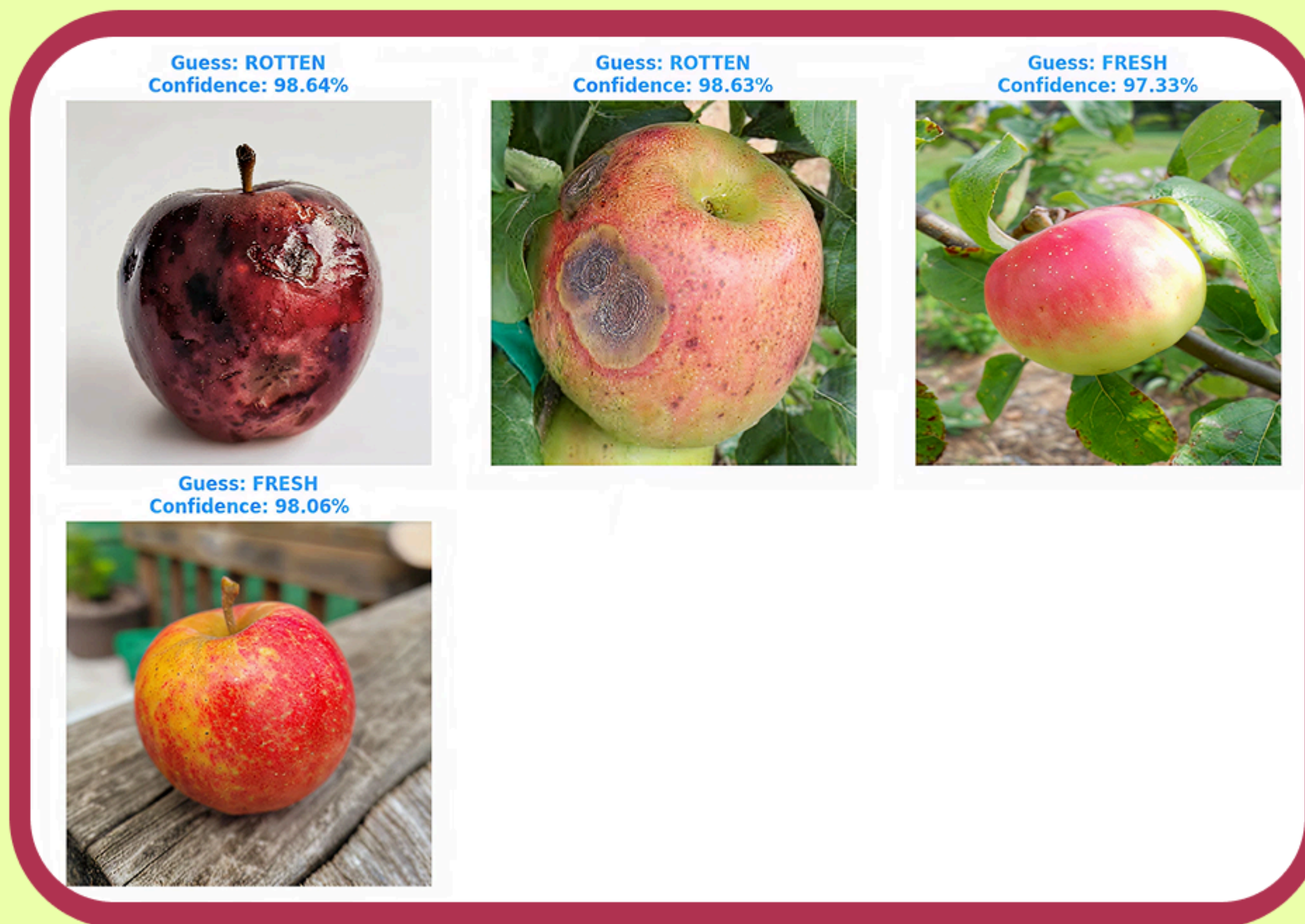
📊 Classification Report (Custom CNN):

	precision	recall	f1-score	support
Fresh	0.92	0.92	0.92	118
Rotten	0.94	0.94	0.94	155
accuracy			0.93	273
macro avg	0.93	0.93	0.93	273
weighted avg	0.93	0.93	0.93	273

Visual Sanity Check



Real-World Testing



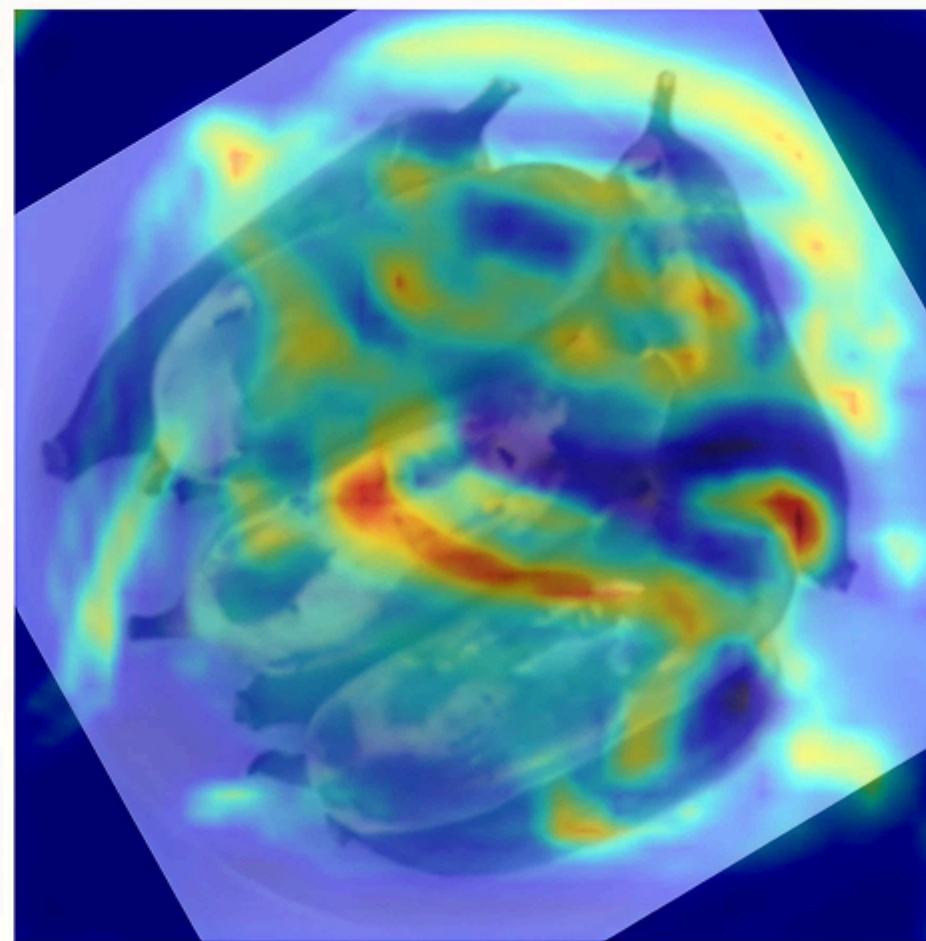
Grad-CAM Visualization



Original Image
True Label in Dataset: ROTTEN
AI Guess: ROTTEN



Grad-CAM Heatmap
(Red/Yellow = Where AI Looked)





KEEP IT FRESH!
THANK YOU!

Presented by:

Alorro ● Bocala ● Dañocup ● Maravilla ● Senibalo

